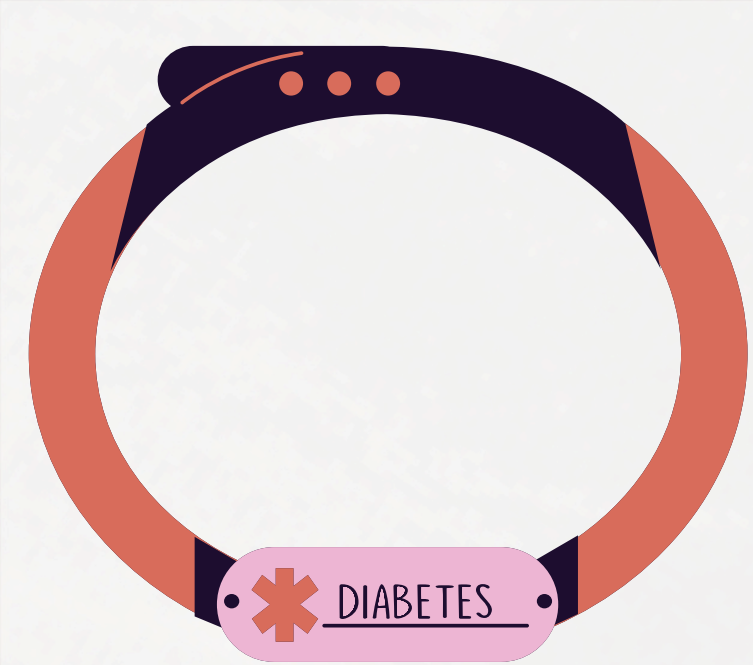




Symptom-based Diabetes Classification & Unsupervised Type Profiling

Dual-Stage Machine Learning Framework for
Early Clinical Triage

STUDENT
Filippo Velli



Problem Statement

Context

Diabetes → pervasive global challenge, causing severe long-term complications like neuropathy and cardiovascular disease

- **Type 1**, which requires acute intervention due to autoimmune beta-cell destruction
- **Type 2**, which is driven by progressive insulin resistance

Gap

Difficulty in **distinguishing** at the beginning which type a patient has

Risk of patients may not be funneled into the distinct therapeutic pathways required for their specific clinical presentation or prevention procedures

Objective

Correct identification of diabetic patients by using type specific signatures

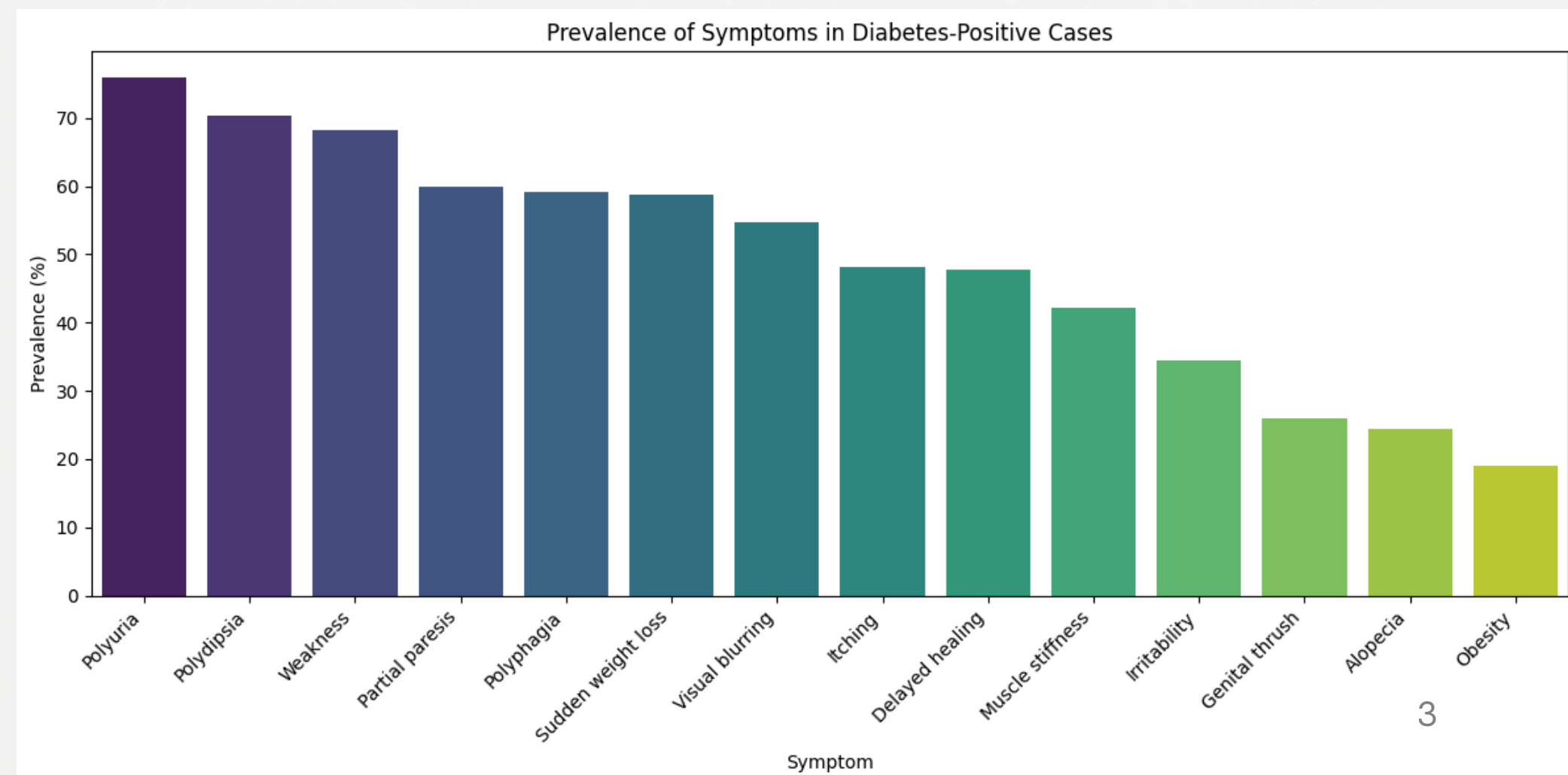
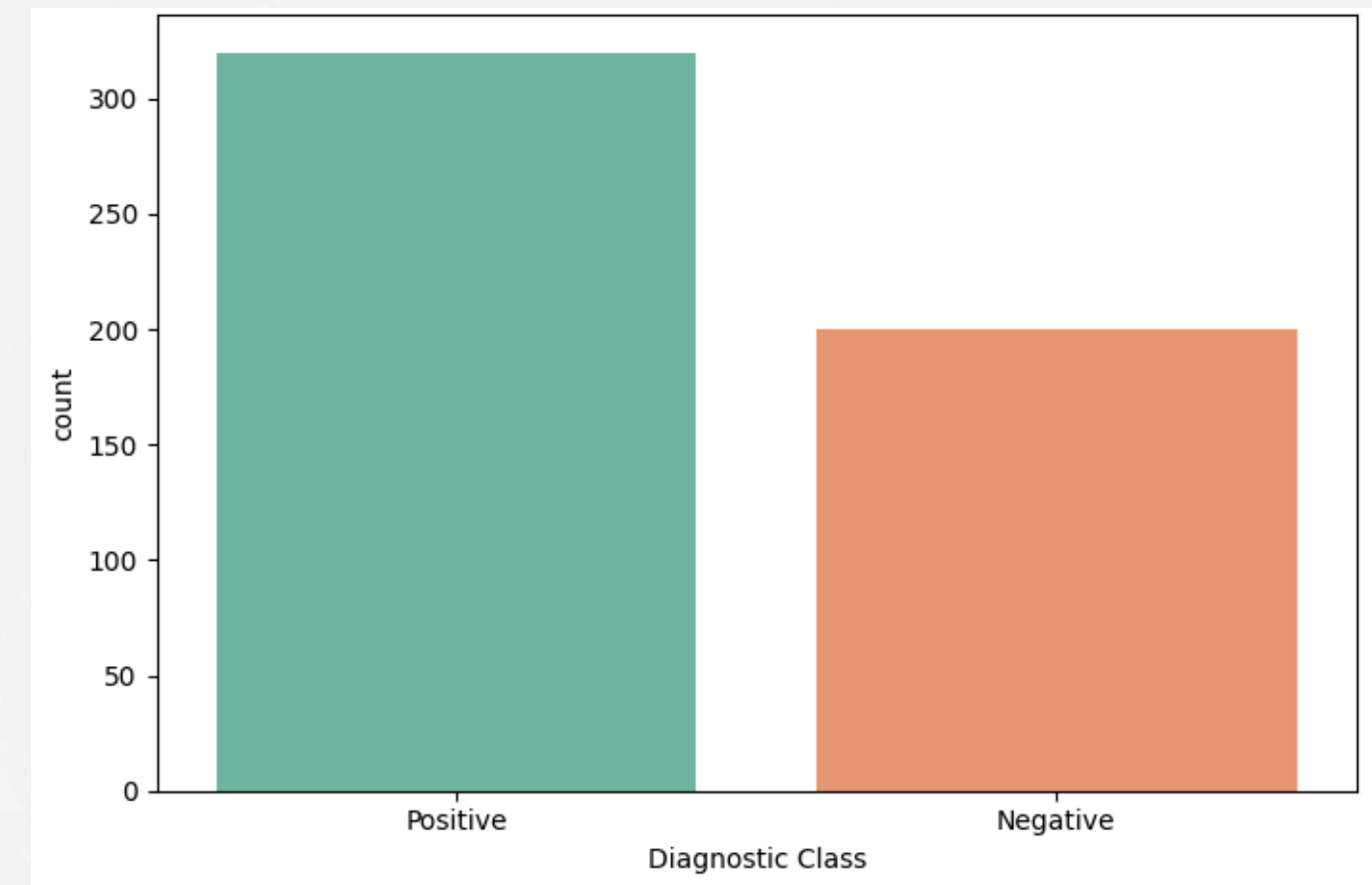
Ensuring more precise and timely clinical intervention and prevention

Dataset Overview

Dataset originating from the **Sylhet Diabetes Hospital**

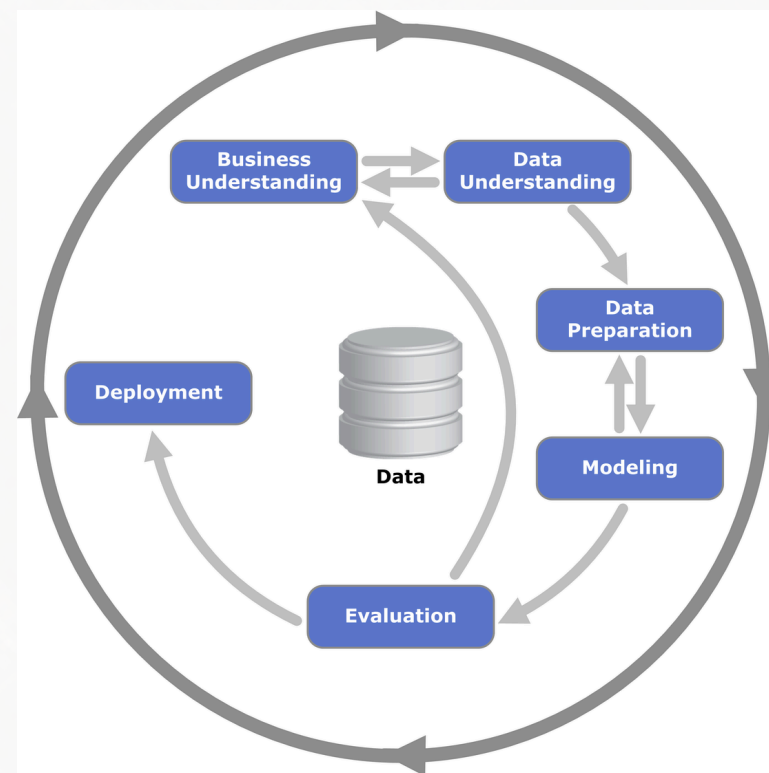
Characteristics

- 520 patients
- Binary target: Positive/Negative
- 58% positivity
- 16 features
- fully complete with no null values
- based on self-reports



Methodology: Dual-Stage Framework

CRISP-MD works as a starting point



FIRST STAGE - SUPERVISED CLASSIFICATION

Establishment of an initial screening pipeline using ensemble methods to separate healthy subjects from the diabetic cohort accurately

SECOND STAGE - UNSUPERVISED CLUSTERING

Application of centroid-based clustering to the results of stage 1 to isolate patients affected by different types of diabetes

Thank you

STUDENT
Filippo Velli